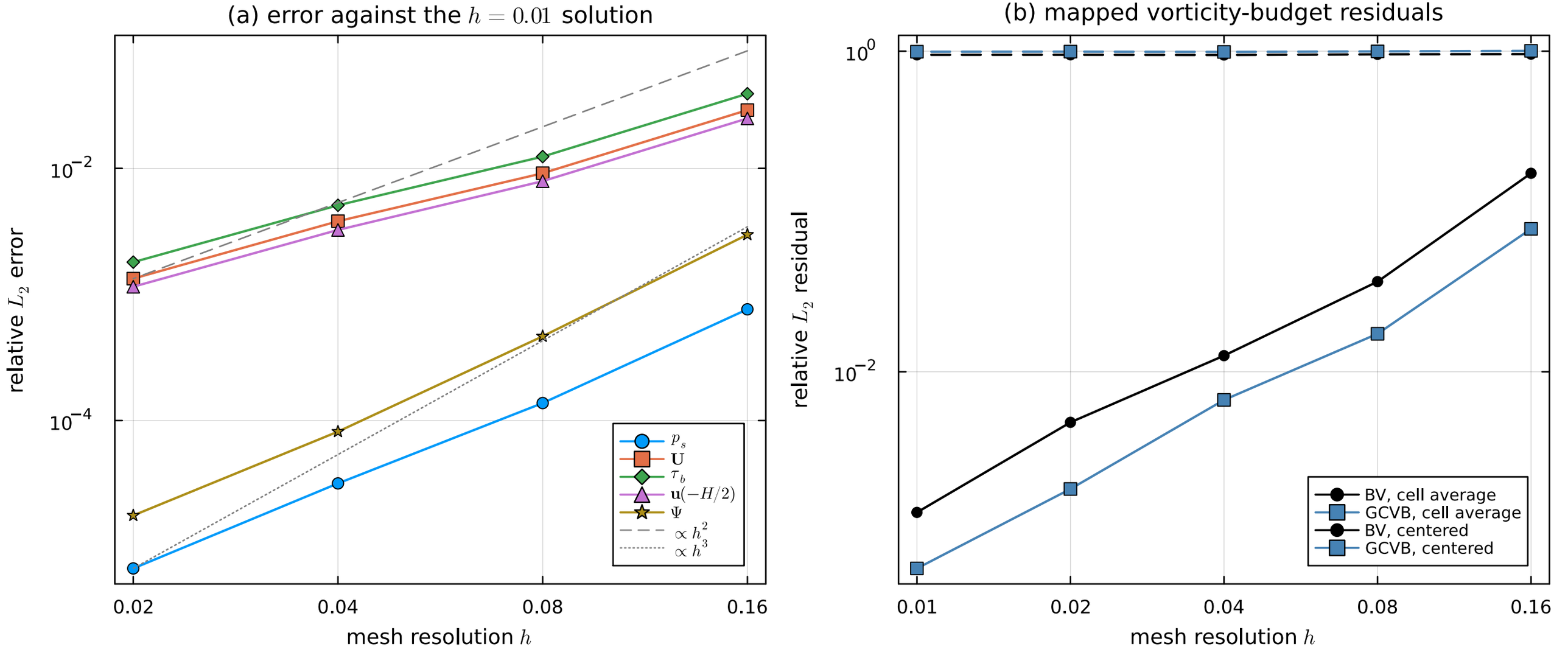
(a) error against the $h = 0.01$ solution
relative L_2 error
10^{-2}
10^{-4}
0.02
0.04
0.08
0.16
mesh resolution h
p_s
$\mathbf{U}$
τ_b
$\mathbf{u}(-H/2)$
Ψ
$\propto h^2$
$\propto h^3$
(b) mapped vorticity-budget residuals
relative L_2 residual
10^0
10^{-2}
0.01
0.02
0.04
0.08
0.16
mesh resolution h
BV, cell average
GCVB, cell average
BV, centered
GCVB, centered